The second Kahn–Kalai conjecture

Rewritten for readability from the OpenAI manuscript of the same title

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1 The result

Graphs are finite and simple. $N(G,F)$ is the number of subgraphs of G isomorphic to F (ordinary copies: G may have further edges among the chosen vertices). $G(n,p)$ is the random graph on $[n]=\{1,\dots,n\}$ in which each edge is present independently with probability p. For a graph H with at least one edge and at most n vertices, define

$$p_{\mathrm{c}}(n,H)=\inf\{p\in[0,1]:\mathbb{P}(N(G(n,p),H)>0)\ge\tfrac12\},$$
$$p_{\mathrm{E}}(n,H)=\inf\{p\in[0,1]:\mathbb{E}N(G(n,p),F)\ge\tfrac12 \text{ for every subgraph } F\subseteq H\},$$

the *containment threshold* and the *graph expectation threshold*. Three elementary facts:

(a) $\mathbb{E}N(G(n,p),F)=N(K_n,F)\,p^{e(F)}$, since each copy of F in K_n is present with probability $p^{e(F)}$. This expectation and the probability defining p_{c} are polynomials in p, and every constraint holds at $p=1$; hence both infima are attained.

(b) $p_{\mathrm{E}}(n,H)>0$: the single-edge subgraph requires $\binom{n}{2}p\ge\frac12$.

(c) $p_{\mathrm{E}}(n,H)\le p_{\mathrm{c}}(n,H)$: a graph containing H contains every $F\subseteq H$, so at $p=p_{\mathrm{c}}(n,H)$ every F satisfies $\mathbb{E}N(G(n,p),F)\ge\mathbb{P}(N(G(n,p),F)>0)\ge\frac12$.

Theorem 1.1. *For every $n\ge2$ and every finite simple graph H with $h=e(H)\ge1$ edges and at most n vertices,*

$$p_{\mathrm{c}}(n,H)\le\min\{1,\ 2048\mathrm{e}^{50}p_{\mathrm{E}}(n,H)(1+\log_2 h)\}. \tag{1.1}$$

In particular, $p_{\mathrm{c}}(n,H)\le 6144\mathrm{e}^{50}p_{\mathrm{E}}(n,H)\log_2 n$.

The constants are universal and H may vary with n. Kahn and Kalai [8] use expected count 1 instead of $\frac12$; by (a) this multiplies the critical density of a subgraph with $x\ge1$ edges by $2^{1/x}\le2$, so it changes p_{E} by a factor in $[1,2]$ and gives the same conjecture.

Conventions. e is the base of the natural logarithm log. For a finite set X and $\rho\in[0,1]$, μ_ρ is the law of the random subset of X containing each element independently with probability ρ. *Monotonicity* means: if a family of subsets of X is closed under supersets, its μ_ρ-probability is nondecreasing in ρ (couple the two samples so that one contains the other). To keep one meaning per symbol we renamed some of the source's letters: K,i (source: R,r) for intersection sizes in Lemma 2.1, τ (source: t) for the number of successes in Section 3, x (source: r) for $|S|$ in Section 4, and κ for the transition probabilities, which the source leaves unnamed.

Proof at a glance

Key idea. A copy of H is grown in stages along a chain of edge sets $\varnothing = H_0 \subset \cdots \subset H_k = H$. All growth histories form a tree; each arc is labelled by the edges added at that stage. One shows that a random edge set contains all labels along some root-to-leaf path. The same random set is reused at all stages, which is affordable because labels along one history are disjoint.

1. **Lemma 2.1 (one family).** Call a random set A *a-spread* if $\mathbb{P}(J \subseteq A) \le a^{|J|}$ for all nonempty J. Given an a-spread family of sets of size at most m and $W \sim \mu_\rho$ with $\rho \ge \mathrm{e}^{50}a$, a randomized rule selects a member A_b whose *fragment* $A_b \setminus W$ has size at most $m/2$, except with probability e^{-9m}. A change-of-measure bound says that selecting b distorts the law of W outside A_b by at most a factor e^{11m}.

2. **Lemma 3.2 (one reduction)** applies Lemma 2.1 at every node of a tree with one common W. With probability at least $3/4$ it returns a subtree labelled by fragments: all label sizes are halved, spread worsens by a factor close to 1, and W together with a path of the new tree covers a path of the old one. Disjointness along paths is what makes the change-of-measure bound applicable down the tree.

3. **Theorem 3.1 (tree covering)** iterates Lemma 3.2 with fresh sets. After $\lfloor \log_2 \ell_k \rfloor + 1$ successes all labels are empty, so the union of the sets used contains a path of the original tree. This is the origin of the factor $1 + \log_2 h$.

4. **Lemma 4.1 (graph hierarchy)**, independent of items 1–3, builds the chain and the tree for H: every stage is $128p_{\mathrm{E}}$-spread and label sizes grow by a factor at least 16.

5. **Theorem 1.1** is Theorem 3.1 applied to the tree of Lemma 4.1; note $16 \cdot 128 = 2048$.

Dependencies: Lemma 2.1 $\Rightarrow$ Lemma 3.2 $\Rightarrow$ Theorem 3.1; Theorem 3.1 and Lemma 4.1 $\Rightarrow$ Theorem 1.1.

Context (quoted from the source; not used in the proof)

Thresholds for fixed targets are classical [5, 2]; here H may grow with n. For a perfect matching on n vertices, $p_{\mathrm{E}} = \Theta(1/n)$ but $p_{\mathrm{c}} = \Theta(\log n/n)$ [6, 8], and Hamilton cycles show the same gap [8], so a logarithm is needed; other spanning targets, such as the cube [14], have required different methods.

For an increasing family $\mathcal{F} \subseteq 2^X$, a *cover* is a family $\mathcal{G}$ such that every member of $\mathcal{F}$ contains a member of $\mathcal{G}$; its cost at p is $\sum_{S\in\mathcal{G}} p^{|S|}$, and $q(\mathcal{F})$ is the largest p at which some cover costs at most $\frac{1}{2}$. Fractional covers define $q_f(\mathcal{F}) \ge q(\mathcal{F})$ similarly [15, Section 6]. The bound $p_{\mathrm{c}}(\mathcal{F}) = O(q_f(\mathcal{F}) \log \ell)$ was proved in [7], using the fragment method of [1, 9], and the same bound with q (the abstract Kahn–Kalai conjecture) in [13]; q and q_f agree up to a constant factor by [12, Theorem 1.1]. For the family $\mathcal{F}_H$ of edge sets containing a copy of H, these results alone do not give Theorem 1.1, because $p_{\mathrm{E}}(n, H) \le q(\mathcal{F}_H)$. Reason: by (a), p_{E} is the largest, over subgraphs $F \subseteq H$ with an edge, of the density at which $\mathbb{E}N(G(n,p), F) = \frac{1}{2}$; the copies of a maximizing F form a cover of $\mathcal{F}_H$ whose cost at p is $\mathbb{E}N(G(n,p), F)$, which equals $\frac{1}{2}$ at $p = p_{\mathrm{E}}$.

Mossel, Niles-Weed, Sun and Zadik [11, Theorem 1] proved a single-logarithm bound relative to a modified threshold (expected count at least $N(H,F)/2$), which equals $q_f(\mathcal{F}_H)$ [3, Equation (1)]. Dubroff, Kahn and Park proved $p_{\mathrm{c}} = O(p_{\mathrm{E}} \log^3 n)$, and the single logarithm when $p_{\mathrm{E}} < 1/(3n)$ [3, Theorems 1.2–1.5]; they also compared $N(H,F)$ with expected counts when F is a clique, a cycle or a bounded-degree tree [4, Theorems 1.4–1.5]. Tran [16] obtained $q_f(\mathcal{F}_H) = O(p_{\mathrm{E}} \log(2e(H)))$,

hence a loss $O(\log^2(2e(H)))$ in general, proved the conjecture for trees and for graphs whose average degree is at least the logarithm of their maximum degree, and proposed a coupling conjecture [16, Section 4] under which the logarithmic sampling cost would be paid only once.

What is new. Lemma 4.1 follows Tran's maximization argument [16, Section 3] (antecedents: [9, Claim 29], [1, Lemma 2.4]); in the form proved below it gives spread $128p_{\mathrm{E}}$ with no logarithm. The transition of Section 2 is from [10]. The new step is Theorem 3.1: all stages are reduced with the same random sets, so the logarithm is paid once for the whole chain.

2 Resampling one spread family

Throughout this section X is a finite set and $(A_b)_{b\in B}$ is a finite indexed family of subsets of X (repetitions and empty sets allowed) with weights $\nu_b > 0$, $\sum_b \nu_b = 1$. For $a > 0$ the family is *a-spread* if

$$\mathbb{P}_{c\sim\nu}(J \subseteq A_c) \le a^{|J|} \qquad (\varnothing \ne J \subseteq X). \tag{2.1}$$

Fix $0 < \rho < 1$. For $Y \subseteq X$ and $b \in B$ put

$$Z(Y) = \sum_{d\in B} \nu_d \rho^{-|A_d|} \mathbf{1}_{\{A_d\subseteq Y\}}, \qquad \kappa(b \mid Y) = \frac{\nu_b \rho^{-|A_b|}\mathbf{1}_{\{A_b\subseteq Y\}}}{Z(Y)}. \tag{2.2}$$

The transition. This is the planted-set and posterior-resampling construction of [10, Sections 2.1–2.3]. Given a set $W \subseteq X$:

1. draw a *source* c with law ν and set $Y = W \cup A_c$;
2. given (W, c), draw a *target* b with probability $\kappa(b \mid Y)$;
3. output the *fragment* $T = A_b \setminus W$.

Step 2 makes sense because $Z(Y) \ge \nu_c \rho^{-|A_c|} > 0$. Thus, for a fixed W,

$$\mathbb{P}(\text{source} = c,\ \text{target} = b \mid W) = \nu_c\, \kappa(b \mid W \cup A_c). \tag{2.3}$$

For every W the source has law ν; the target in general does not. Three observations:

(O1) *Inclusions.* If the pair (c, b) has positive probability, then $A_b \subseteq W \cup A_c$, hence

$$T \subseteq A_c \cap A_b, \qquad A_b \subseteq W \cup T. \tag{2.4}$$

The first inclusion compares the fragment with the source. The second says that W and the fragment together cover the target label.

(O2) *Locality.* Since $\mathbf{1}_{\{A_d\subseteq W\cup A_c\}} = \mathbf{1}_{\{A_d\setminus A_c\subseteq W\}}$, the probabilities (2.3), the value $Z(Y)$ and the fragment depend on W only through $W \cap \bigcup_{d\in B} A_d$.

(O3) *Law of Y.* Now let $W \sim \mu_\rho$, independently of c. For fixed c and $y \supseteq A_c$, we have $W \cup A_c = y$ exactly when W contains $y \setminus A_c$ and misses $X \setminus y$ (elements of A_c are unconstrained), so

$$\mathbb{P}(W \cup A_c = y) = \rho^{|y|-|A_c|}(1-\rho)^{|X\setminus y|} = \mu_\rho(y)\rho^{-|A_c|};$$

the probability is 0 if $A_c \not\subseteq y$. Averaging over $c \sim \nu$,

$$\mathbb{P}(Y = y) = \mu_\rho(y) Z(y) \qquad (y \subseteq X). \tag{2.5}$$

(By Bayes' formula, $\kappa(\cdot \mid Y)$ is then the conditional law of the source given Y, so the target is a conditionally independent resample of the source. This motivates the rule but is not used below.)

Lemma 2.1 (Local resampling)**.** *Suppose $|A_b| \le m$ for all b, where m is a positive integer, the family is a-spread, and*

$$0 < \rho < 1, \qquad a/\rho \le \mathrm{e}^{-50}.$$

Let $W \sim \mu_\rho$ be independent of the source and perform the transition; probabilities and expectations below are over W, source and target jointly. Call the outcome a local failure *if $Z(Y) < \mathrm{e}^{-10m}$ or $|T| > m/2$. Then*

$$\mathbb{P}(\textit{local failure}) \le \mathrm{e}^{-9m}. \tag{2.6}$$

Moreover, for each $b \in B$ and every function $f \ge 0$ of W that depends only on $W \setminus A_b$,

$$\mathbb{E}\big[\mathbf{1}_{\{\text{target}=b,\ Z(Y)\ge \mathrm{e}^{-10m}\}} f(W)\big] \le \mathrm{e}^{11m}\nu_b\,\mathbb{E}f(W). \tag{2.7}$$

The lemma is used at every node of the tree in Lemma 3.2: (2.6) bounds the failure mass, and (2.7) is applied with f a potential of the subtree below b.

Proof. Plan. Small Z is rare by (2.5). When Z is not small, κ is at most $\mathrm{e}^{10m}\nu_b\rho^{-|A_b|}$ times an indicator, and everything reduces to the intersection sizes $|A_c \cap A_b|$, which spread controls.

Step 1 (small denominator). By (2.5),

$$\mathbb{P}(Z(Y) < \mathrm{e}^{-10m}) = \sum_y \mu_\rho(y) Z(y) \mathbf{1}_{\{Z(y)<\mathrm{e}^{-10m}\}} \le \mathrm{e}^{-10m}.$$

Step 2 (intersections). Fix b and let $K = |A_c \cap A_b|$ with $c \sim \nu$. If $K = i$, then $J = A_c \cap A_b$ is an i-element subset of A_b with $J \subseteq A_c$. There are at most $\binom{m}{i}$ such J, each with probability at most a^i by (2.1) (trivially if $i = 0$). Hence, for every set $\mathcal{I} \subseteq \{0, \dots, m\}$,

$$\sum_c \nu_c \rho^{-|A_c\cap A_b|} \mathbf{1}_{\{|A_c\cap A_b|\in\mathcal{I}\}} = \sum_{i\in\mathcal{I}} \mathbb{P}(K = i)\rho^{-i} \le \sum_{i\in\mathcal{I}} \binom{m}{i} (a/\rho)^i. \tag{2.8}$$

Step 3 (cancellation). For fixed c, b, the event $A_b \subseteq W \cup A_c$ is the event $A_b \setminus A_c \subseteq W$, of probability $\rho^{|A_b\setminus A_c|}$. Therefore

$$\rho^{-|A_b|}\,\mathbb{P}_W(A_b \subseteq W \cup A_c) = \rho^{-|A_c\cap A_b|}. \tag{2.9}$$

Step 4 (large fragment). On $\{Z(Y) \ge \mathrm{e}^{-10m}\}$ we have $\kappa(b \mid Y) \le \mathrm{e}^{10m}\nu_b\rho^{-|A_b|}\mathbf{1}_{\{A_b\subseteq Y\}}$, and by (2.4) $|T| > m/2$ forces $|A_c \cap A_b| > m/2$. Summing (2.3) over pairs and averaging over W,

$$\begin{aligned}
\mathbb{P}\big(Z(Y) \ge \mathrm{e}^{-10m},\ |T| > m/2\big) &\le \mathrm{e}^{10m} \sum_{c,b} \nu_c\nu_b\rho^{-|A_b|}\mathbb{P}_W(A_b \subseteq W \cup A_c)\mathbf{1}_{\{|A_c\cap A_b|>m/2\}} \\
&= \mathrm{e}^{10m} \sum_b \nu_b \sum_c \nu_c \rho^{-|A_c\cap A_b|}\mathbf{1}_{\{|A_c\cap A_b|>m/2\}} && \text{by (2.9)} \\
&\le \mathrm{e}^{10m} \sum_{i>m/2} \binom{m}{i}(a/\rho)^i && \text{by (2.8)} \\
&\le \mathrm{e}^{10m} 2^m \mathrm{e}^{-25m},
\end{aligned}$$

since $(a/\rho)^i \le \mathrm{e}^{-50i} \le \mathrm{e}^{-25m}$ for $i > m/2$ and $\sum_i \binom{m}{i} = 2^m$.

Step 5 (total). Steps 1 and 4 give $\mathbb{P}(\text{local failure}) \le \mathrm{e}^{-10m} + \mathrm{e}^{-(15-\log 2)m}$. Dividing by e^{-9m}, this is at most $\mathrm{e}^{-m} + \mathrm{e}^{-(6-\log 2)m} \le \mathrm{e}^{-1} + \mathrm{e}^{-5} < 1$ because $m \ge 1$. This proves (2.6).

Step 6 (change of measure). Bounding κ as in Step 4,

$$\mathbb{E}\big[\mathbf{1}_{\{\text{target}=b,\ Z(Y)\geq \mathrm{e}^{-10m}\}} f(W)\big] \leq \mathrm{e}^{10m}\nu_b \sum_c \nu_c \rho^{-|A_b|}\mathbb{E}\big[\mathbf{1}_{\{A_b\setminus A_c\subseteq W\}} f(W)\big].$$

The indicator depends only on $W \cap A_b$ and $f(W)$ only on $W \setminus A_b$; under the product measure μ_ρ these are independent, so the last expectation equals $\rho^{|A_b\setminus A_c|}\mathbb{E}f(W)$. By the arithmetic of (2.9) and then (2.8) with $\mathcal{I} = \{0,\dots,m\}$, the right-hand side is at most

$$\mathrm{e}^{10m}\nu_b\,\mathbb{E}f(W)\sum_c \nu_c\rho^{-|A_c\cap A_b|} \leq \mathrm{e}^{10m}\nu_b\,\mathbb{E}f(W)\,(1+a/\rho)^m \leq \mathrm{e}^{11m}\nu_b\,\mathbb{E}f(W),$$

using $1 + a/\rho \leq \mathrm{e}$. □

3 Covering a tree

Definitions. A *probability tree on X with $k \geq 0$ levels* is a finite rooted tree whose leaves all have depth k (the root has depth 0), together with, for each non-leaf node v, a probability distribution ν_v with positive weights on its set of children $\mathrm{Ch}(v)$, and a label $A_v(b) \subseteq X$ on the arc from v to each child b. *Level j* $(1 \leq j \leq k)$ is the set of arcs from depth $j-1$ to depth j; a node at depth $j-1$ has *outgoing level j*. Different arcs may carry equal labels; nodes are never identified. A *path P* always runs from the root to a leaf, and $\mathcal{U}(P) = \bigcup_{(v,b)\in P} A_v(b)$. If $k = 0$ the only path is empty, with $\mathcal{U} = \varnothing$.

The tree is *disjoint* if the labels along each path are pairwise disjoint. Level j has *capacity m* if all its labels have size at most m, and is *a-spread* if, for every node v with outgoing level j, the family $(A_v(b))_{b\in\mathrm{Ch}(v)}$ with weights ν_v is a-spread in the sense of (2.1). Nothing is assumed about the law of a whole path.

Theorem 3.1 (Tree covering)**.** *Let $\mathcal{T}$ be a disjoint probability tree on a finite set X, with $k \geq 1$ levels. Suppose that every level is σ-spread, where $\sigma > 0$, and that the levels have positive integer capacities $\ell_1,\dots,\ell_k$ with*

$$\ell_{i+1} \geq 16\ell_i \qquad (1 \leq i < k).$$

Set $r = \min\{1,\ 16\mathrm{e}^{50}\sigma(1+\log_2 \ell_k)\}$. Then

$$\mathbb{P}_{W\sim\mu_r}\big(\mathcal{U}(P) \subseteq W \text{ for some path } P \text{ of } \mathcal{T}\big) \geq \tfrac{2}{3}.$$

It is used once, in Section 4, with $\sigma = 128p_{\mathrm{E}}(n,H)$ and $\ell_k \leq e(H)$.

3.1 One simultaneous reduction

Lemma 3.2 (Reduction)**.** *Let $\mathcal{T}$ be a disjoint probability tree with $k \geq 1$ levels and positive integer capacities $m_1,\dots,m_k$ satisfying $m_{j+1} \geq 16m_j$. Suppose level j is α_j-spread, where*

$$0 < \alpha_j \leq a, \qquad \rho = \mathrm{e}^{50}a < 1.$$

There is a deterministic procedure which, given $W \sim \mu_\rho$, succeeds with probability at least $3/4$ and on success returns a disjoint probability tree $\mathcal{T}'$ with the same number k of levels, such that:

(i) level j has capacity $\lfloor m_j/2\rfloor$ (possibly 0) and is $\alpha_j/(1-\mathrm{e}^{-m_j})$-spread;

(ii) for every path Q of $\mathcal{T}'$ there is a path P of $\mathcal{T}$ with

$$\mathcal{U}(P) \subseteq W \cup \mathcal{U}(Q). \tag{3.1}$$

It is the single step iterated in the proof of Theorem 3.1.

Proof. Plan. Run the transition of Section 2 at every node with the same W. Discard nodes at which too much probability is lost (Steps 0–1), bound the chance that the root is discarded via a potential (Steps 2–5), and read off the new tree (Steps 6–7).

Step 0 (local quantities). Let v be a node with outgoing level j. Apply the transition to the family $(A_v(b))_{b\in\mathrm{Ch}(v)}$ with weights ν_v and $m = m_j$. Since $\alpha_j \le a$ this family is a-spread, and $a/\rho = \mathrm{e}^{-50}$, so Lemma 2.1 applies. For a fixed W, with probabilities over source and target at v, set

$$d_v(W) = \mathbb{P}(\text{local failure} \mid W), \qquad t_{vb}(W) = \mathbb{P}(\text{no local failure, target} = b \mid W).$$

Then

$$d_v(W) + \sum_{b\in\mathrm{Ch}(v)} t_{vb}(W) = 1, \tag{3.2}$$

and by (O2) both d_v and t_{vb} depend on W only through $W \cap \bigcup_{b\in\mathrm{Ch}(v)} A_v(b)$. These quantities always refer to the input tree, before any pruning.

Step 1 (pruning). Fix W and put $\epsilon_j = \mathrm{e}^{-m_j}$. All leaves are *good*. Proceeding upward, a node v with outgoing level j is good if

$$\sum_{b\in\mathrm{Ch}(v),\ b \text{ good}} t_{vb}(W) \ge 1 - \epsilon_j, \tag{3.3}$$

and *bad* otherwise. By (3.2), at a good node the failure mass plus the mass of bad targets is at most ϵ_j. The procedure *succeeds* when the root is good.

Step 2 (potential). Set $D_v(W) = 0$ at leaves and, for v with outgoing level j,

$$D_v(W) = \epsilon_j^{-1}\Big(d_v(W) + \sum_{b\in\mathrm{Ch}(v)} t_{vb}(W) D_b(W)\Big) \ge 0. \tag{3.4}$$

Claim: every bad node has $D_v(W) > 1$. Induct upward; leaves are good. If v is bad and the claim holds for its bad children, then

$$\epsilon_j D_v \ge d_v + \sum_{b \text{ bad}} t_{vb} D_b \ge d_v + \sum_{b \text{ bad}} t_{vb} = 1 - \sum_{b \text{ good}} t_{vb} > \epsilon_j,$$

by (3.2) and the failure of (3.3).

Step 3 (D_b ignores its incoming label). Let b be a child of v. *Claim: $D_b(W)$ depends on W only through $W \setminus A_v(b)$.* By (3.4), D_b is a function of the quantities d_w, t_{wc} for nodes w in the subtree rooted at b. By Step 0 these depend on W only on the labels $A_w(c)$ of arcs of that subtree. Each such arc lies on a root-to-leaf path through the arc (v, b), so by disjointness its label is disjoint from $A_v(b)$.

Step 4 (averaging over W). Now let $W \sim \mu_\rho$, and let v have outgoing level j. By (2.6), $\mathbb{E}d_v(W) \le \mathrm{e}^{-9m_j}$. For a child b, absence of local failure implies $Z(Y) \ge \mathrm{e}^{-10m_j}$, so

$$\mathbb{E}\big[t_{vb}(W) D_b(W)\big] \le \mathbb{E}\big[\mathbf{1}_{\{\text{target}=b,\ Z(Y)\ge \mathrm{e}^{-10m_j}\}} D_b(W)\big] \le \mathrm{e}^{11m_j}\nu_v(b)\,\mathbb{E}D_b(W);$$

the last step is (2.7) with $f = D_b$, admissible by Step 3. Inserting both bounds into (3.4), with $\epsilon_j^{-1} = \mathrm{e}^{m_j}$,

$$\mathbb{E}D_v(W) \le \mathrm{e}^{-8m_j} + \mathrm{e}^{12m_j} \sum_{b\in\mathrm{Ch}(v)} \nu_v(b)\,\mathbb{E}D_b(W). \tag{3.5}$$

This is the step that controls the dependence created by reusing W.

Step 5 (the root is rarely bad). *Claim:* for v with outgoing level j,

$$\mathbb{E}D_v(W) \le \sum_{i=j}^{k} \exp\Big(12\sum_{l=j}^{i-1} m_l - 8m_i\Big). \tag{3.6}$$

Induct upward. For leaves (read $j = k+1$) both sides are 0. For the inductive step, children have outgoing level $j+1$; insert their bound into (3.5) and use $\sum_b \nu_v(b) = 1$: the term e^{-8m_j} is the summand $i = j$, and e^{12m_j} times the i-th summand for $j+1$ is the i-th summand for j.

For the root o ($j = 1$): since $m_l \le 16^{-(i-l)}m_i$, we get $\sum_{l<i} m_l \le m_i/15$, so each exponent is at most $(12/15 - 8)m_i \le -7m_i$. The m_i are distinct positive integers, so by Step 2 and Markov's inequality

$$\mathbb{P}(o \text{ is bad}) \le \mathbb{P}(D_o(W) > 1) \le \mathbb{E}D_o(W) \le \sum_{m\ge 1} \mathrm{e}^{-7m} < \tfrac{1}{4}.$$

Step 6 (the reduced tree). Assume the root is good. Start from the root; at each retained non-leaf node v (necessarily good), with outgoing level j, let

$$\gamma_v(W) = \sum_{b \text{ good}} t_{vb}(W) \ge 1 - \epsilon_j > 0.$$

Retain exactly the good children b with $t_{vb}(W) > 0$, give b the weight $t_{vb}(W)/\gamma_v(W)$, and relabel the arc (v, b) by $T_b = A_v(b) \setminus W$. Checks:

- *Probability tree with k levels.* Weights are positive and sum to 1. Every retained non-leaf node has a retained child, since $\gamma_v > 0$, and that child is good; so all leaves of $\mathcal{T}'$ are leaves of $\mathcal{T}$.
- *Disjoint.* $T_b \subseteq A_v(b)$.
- *(ii).* A path Q of $\mathcal{T}'$ passes through the nodes of a path P of $\mathcal{T}$, and $A_v(b) \subseteq W \cup T_b$ on each arc.
- *Capacity.* T_b is the fragment of any transition at v with target b. If $t_{vb}(W) > 0$, some such transition is not a local failure, so $|T_b| \le m_j/2$, hence $|T_b| \le \lfloor m_j/2 \rfloor$.

Step 7 (spread). Fix a retained node v with outgoing level j and a nonempty $J \subseteq X$. In the transition at v for the fixed W, let E_v be the event "no local failure and the target is good", so $\mathbb{P}(E_v \mid W) = \gamma_v(W)$. The new child law is the law of the target conditioned on E_v, and the new label is the fragment T. By (2.4), $T \subseteq A_v(c)$ for the source c, and c has law ν_v whatever W is. Hence

$$\mathbb{P}(J \subseteq T \mid E_v, W) = \frac{\mathbb{P}(J \subseteq T,\ E_v \mid W)}{\gamma_v(W)} \le \frac{\mathbb{P}_{c\sim\nu_v}(J \subseteq A_v(c))}{1-\epsilon_j} \le \frac{\alpha_j^{|J|}}{1-\epsilon_j} \le \Big(\frac{\alpha_j}{1-\epsilon_j}\Big)^{|J|},$$

the last step because $|J| \ge 1$. This proves (i).

The construction uses only the numbers $t_{vb}(W)$, so it is deterministic given W; sources and targets serve only the analysis. □

3.2 Iteration

Proof of Theorem 3.1. Plan. Apply Lemma 3.2 repeatedly with fresh sets, deleting levels once their labels are empty; check that the hypotheses persist; count the attempts.

Step 1 (parameters). If $r = 1$ then $W = X$ and the claim is trivial. Assume $r < 1$ and set

$$a = 4\sigma, \qquad \rho = \mathrm{e}^{50}a, \qquad s = \lfloor \log_2 \ell_k \rfloor + 1.$$

Since $\ell_k \geq 1$, $r \geq 16\mathrm{e}^{50}\sigma = 4\rho$, so $0 < \rho < 1$.

Step 2 (procedure). Draw independent sets $W_1, W_2, \ldots$ of law μ_ρ. Keep a current tree $\mathcal{T}^{(\tau)}$, where τ counts successes and $\mathcal{T}^{(0)} = \mathcal{T}$. An *attempt* applies Lemma 3.2 to the current tree with the next set. On failure the tree is unchanged. On success let $\mathcal{T}'$ be the output and *trim* it: capacities increase with the level, so the levels of capacity 0 form an initial segment; follow one downward path from the root through these levels (chosen by a fixed rule) to a node v^*, and let $\mathcal{T}^{(\tau+1)}$ be the subtree of $\mathcal{T}'$ rooted at v^*. Stop when the current tree has no levels. Lemma 3.2 itself deletes nothing; levels disappear only by trimming. Trimming is harmless:

- Each path Q of $\mathcal{T}^{(\tau+1)}$, preceded by the chosen prefix, is a path of $\mathcal{T}'$ with the same label union, because the prefix labels are empty. So by (3.1), if W denotes the set used,

$$\text{for every path } Q \text{ of } \mathcal{T}^{(\tau+1)} \text{ there is a path } P \text{ of } \mathcal{T}^{(\tau)} \text{ with } \mathcal{U}(P) \subseteq W \cup \mathcal{U}(Q). \tag{3.7}$$

- Nodes of $\mathcal{T}^{(\tau+1)}$ keep their children, child laws and labels from $\mathcal{T}'$. So the tree is a disjoint probability tree, and the capacity and spread bounds of Lemma 3.2(i) persist on the remaining levels.

Step 3 (invariants). Index levels by their index i in $\mathcal{T}$. After τ successes, by induction and $\lfloor \lfloor y \rfloor / 2 \rfloor = \lfloor y/2 \rfloor$:

- level i remains if and only if $m_i(\tau) = \lfloor \ell_i / 2^\tau \rfloor \geq 1$, and then has capacity $m_i(\tau)$;
- remaining capacities satisfy $m_{i+1}(\tau) \geq \lfloor 16\ell_i / 2^\tau \rfloor \geq 16 m_i(\tau)$;
- a remaining level i is $\alpha_i(\tau)$-spread, where $\alpha_i(\tau) = \sigma \prod_{u=0}^{\tau-1} (1 - \mathrm{e}^{-m_i(u)})^{-1}$, by Lemma 3.2(i) applied at each earlier success.

Step 4 ($\alpha_i(\tau) \leq a$). For a remaining level, $m_i(0) > m_i(1) > \cdots > m_i(\tau - 1) \geq 1$, because $\lfloor y/2 \rfloor < y$ for integers $y \geq 1$. So the factors of $\alpha_i(\tau)$ belong to distinct positive integers m, and $\alpha_i(\tau) \leq \sigma \prod_{m \geq 1} (1 - \mathrm{e}^{-m})^{-1}$. Using $-\log(1 - z) \leq z/(1 - z) \leq z/(1 - \mathrm{e}^{-1})$ for $0 < z \leq \mathrm{e}^{-1}$,

$$\sum_{m \geq 1} -\log(1 - \mathrm{e}^{-m}) \leq \frac{1}{1 - \mathrm{e}^{-1}} \sum_{m \geq 1} \mathrm{e}^{-m} = \frac{\mathrm{e}^{-1}}{(1 - \mathrm{e}^{-1})^2} < 0.93 < \log 4,$$

so the product is below 4 and $\alpha_i(\tau) \leq 4\sigma = a$. By Steps 1–4, as long as it has a level, the current tree satisfies the hypotheses of Lemma 3.2 with these a, ρ. The current tree is determined by the earlier sets and the next set is independent of them; so each attempt succeeds with conditional probability at least 3/4 given the past.

Step 5 (termination and recovery). Level k remains exactly while $2^\tau \leq \ell_k$, that is $\tau \leq s - 1$. So the procedure stops after exactly s successes, and $\mathcal{T}^{(s)}$ is a single node whose only path Q_s is empty. Let $W^{(1)}, \ldots, W^{(s)}$ be the sets used at the successes. Applying (3.7) for $\tau = s, s - 1, \ldots, 1$ gives paths $Q_{\tau-1}$ of $\mathcal{T}^{(\tau-1)}$ with $\mathcal{U}(Q_{\tau-1}) \subseteq W^{(\tau)} \cup \mathcal{U}(Q_\tau)$. Hence

$$\mathcal{U}(Q_0) \subseteq W^{(1)} \cup \cdots \cup W^{(s)}, \qquad Q_0 \text{ a path of } \mathcal{T}.$$

Step 6 (number of attempts). Let G_τ be the number of attempts after the $(\tau-1)$-st success up to and including the τ-th. The event $G_\tau > j$ requires j consecutive failures, each of conditional probability at most $1/4$ given all earlier sets, so $\mathbb{P}(G_\tau > j) \le 4^{-j}$ and $\mathbb{E}G_\tau = \sum_{j\ge 0}\mathbb{P}(G_\tau > j) \le 4/3$. The total number of attempts $N = G_1 + \cdots + G_s$ has $\mathbb{E}N \le 4s/3$, and by Markov's inequality $\mathbb{P}(N > 4s) \le 1/3$.

Step 7 (density). Let $W_* = W_1 \cup \cdots \cup W_{4s}$. It has law μ_{r_*} with

$$r_* = 1-(1-\rho)^{4s} \le 4s\rho = 16\mathrm{e}^{50}\sigma s \le 16\mathrm{e}^{50}\sigma(1+\log_2 \ell_k) = r.$$

On $\{N \le 4s\}$, of probability at least $2/3$, all successful sets are among $W_1, \dots, W_{4s}$, so $\mathcal{U}(Q_0) \subseteq W_*$ by Step 5. The event that a set contains some path union is closed under supersets, so by monotonicity the same bound holds for μ_r. □

4 Application to graph containment

The construction below is the normalized-containment maximization and conditional-extension argument of Tran [16, Section 3, Lemma 3.1], with constants suited to the factor 16; identity (4.2) also appears in [11, Equation (5)].

Conventions. Put $q = p_{\mathrm{E}}(n,H)$, $\sigma = 128q$ and $X = \binom{[n]}{2}$. In this section a graph without isolated vertices is identified with its edge set; H now denotes the edge set of the given graph, and $F \subseteq H$ means an edge subset. A *copy* of such an F is a subset of X isomorphic to F. Let $M(F) = N(K_n, F)$ be the number of copies of F, and $M(I,F) = N(I,F)$ the number of subsets of I isomorphic to F, with $M(\varnothing) = M(I,\varnothing) = 1$. The number q always refers to the original graph. Two remarks justify this:

- If an edge set on $[n]$ contains a copy of the edge set H, the corresponding graph contains the original graph: place its isolated vertices on unused vertices, which exist since the original graph has at most n vertices, and copies need not be induced.
- Each nonempty $F \subseteq H$, on its incident vertices, is a subgraph of the original graph. Since the infimum defining q is attained, fact (a) gives

$$M(F)\, q^{|F|} \ge \tfrac{1}{2}. \tag{4.1}$$

Copy-ratio identity. Let $F \subseteq I \subseteq H$, let $\boldsymbol{I}$ be a uniformly random copy of I, and let F_0 be a fixed copy of F. Then

$$\mathbb{P}(F_0 \subseteq \boldsymbol{I}) = \frac{M(I,F)}{M(F)}. \tag{4.2}$$

Indeed, permutations of $[n]$ preserve the law of $\boldsymbol{I}$ and act transitively on the copies of F, so the left side is the same number π for every F_0. Every value of $\boldsymbol{I}$ contains exactly $M(I,F)$ copies of F; taking expectations in $\sum_{F_0} \mathbf{1}_{\{F_0 \subseteq \boldsymbol{I}\}} = M(I,F)$ gives $M(F)\pi = M(I,F)$. In particular the probability depends only on the isomorphism types of F and I.

Lemma 4.1 (Graph hierarchy)**.** *There is a disjoint probability tree on $X = \binom{[n]}{2}$ whose path unions are copies of H with its isolated vertices removed, such that every level is σ-spread and its label sizes $\ell_1, \dots, \ell_k$ satisfy*

$$1 \le \ell_1, \qquad \ell_{i+1} \ge 16\ell_i, \qquad \ell_k \le e(H).$$

This is the input to Theorem 3.1 in the proof of Theorem 1.1.

Proof. Plan. Choose each $H_{i-1} \subseteq H_i$ by a maximization that forces $|H_{i-1}| < |H_i|/17$ (from first moments) and conditional spread (from maximality).

Step 1 (maximizers are small). For nonempty $I \subseteq H$ and $S \subseteq I$ let

$$R_\sigma(I,S) = \sigma^{-|S|}\,\mathbb{P}(S_0 \subseteq \boldsymbol{I}), \tag{4.3}$$

where S_0 is any copy of S and $\boldsymbol{I}$ a uniform copy of I; by (4.2) this does not depend on S_0. Note $R_\sigma(I,\varnothing) = 1$. *Claim: if $x = |S| \geq |I|/17$, then $R_\sigma(I,S) < 1$.* Here $x \geq 1$. By (4.2), by $1/M(S) \leq 2q^x$ from (4.1), and by $M(I,S) \leq \binom{|I|}{x} \leq (\mathrm{e}|I|/x)^x \leq (17\mathrm{e})^x$,

$$R_\sigma(I,S) \leq \sigma^{-x} \cdot 2q^x(17\mathrm{e})^x = 2\Big(\frac{17\mathrm{e}}{128}\Big)^x \leq \frac{34\mathrm{e}}{128} < 0.73.$$

Hence every maximizer S of $R_\sigma(I,\cdot)$, whose value is at least 1, satisfies $|S| < |I|/17$.

Step 2 (chain). Starting from $I = H$, repeatedly replace the current nonempty I by a maximizer of $R_\sigma(I,\cdot)$ (ties broken by a fixed rule) until $\varnothing$ is reached. In reverse order this gives

$$\varnothing = H_0 \subset H_1 \subset \cdots \subset H_k = H, \qquad H_{i-1} \text{ maximizes } R_\sigma(H_i,\cdot), \qquad |H_{i-1}| < |H_i|/17,$$

with $k \geq 1$.

Step 3 (tree). The root is the empty edge set. A node at depth $i-1$ is a copy S_0 of H_{i-1} together with its history. Its children are all copies I_0 of H_i with $S_0 \subseteq I_0$, with uniform law, and the arc is labelled $I_0 \setminus S_0$. Children exist: take an isomorphism from H_{i-1} to S_0 and extend it injectively to the vertices of H_i, which is possible because H has at most n vertices; the image of H_i is a copy containing S_0. Along a path the copies are nested, so the labels are pairwise disjoint and their union is a copy of H. All leaves have depth k.

Step 4 (sizes). Every label of level i has size $\ell_i = |H_i| - |H_{i-1}| \geq 1$. Moreover $\ell_k \leq |H| = e(H)$ and $\ell_{i+1} > 17|H_i| - |H_i| = 16|H_i| \geq 16\ell_i$.

Step 5 (spread). Fix a node S_0, a copy of H_{i-1}, let A be its random label, and let $\boldsymbol{I}$ be a uniform copy of H_i. Conditioned on $S_0 \subseteq \boldsymbol{I}$, $\boldsymbol{I}$ is uniform on the children of S_0. Let $J \subseteq X$ be nonempty. If $J \cap S_0 \neq \varnothing$ then $\mathbb{P}(J \subseteq A) = 0$. Otherwise

$$\mathbb{P}(J \subseteq A) = \frac{\mathbb{P}(S_0 \cup J \subseteq \boldsymbol{I})}{\mathbb{P}(S_0 \subseteq \boldsymbol{I})}. \tag{4.4}$$

If the numerator is 0 we are done. Otherwise $S_0 \cup J$ lies in some copy I_0 of H_i; an isomorphism $I_0 \to H_i$ maps $S_0 \cup J$ onto an edge subset $U \subseteq H_i$ with $|U| = |S_0| + |J|$, and $S_0 \cup J$ is a copy of U. By (4.3), the numerator is $\sigma^{|S_0|+|J|}R_\sigma(H_i,U)$ and the denominator is $\sigma^{|S_0|}R_\sigma(H_i,H_{i-1})$. Maximality of H_{i-1} gives $R_\sigma(H_i,U) \leq R_\sigma(H_i,H_{i-1})$, hence $\mathbb{P}(J \subseteq A) \leq \sigma^{|J|}$. □

Sanity check. If H is a single edge, then $q = 1/(n(n-1))$, $k = 1$, and the root has $\binom{n}{2}$ equally likely children with one-edge labels; indeed $1/\binom{n}{2} = 2q \leq \sigma$.

Proof of Theorem 1.1. Apply Theorem 3.1 to the tree of Lemma 4.1, with $\sigma = 128q$. Since $\ell_k \leq h$, its density $\min\{1, 16\mathrm{e}^{50} \cdot 128q(1+\log_2 \ell_k)\}$ is at most

$$r = \min\{1,\ 2048\mathrm{e}^{50}q(1+\log_2 h)\}.$$

The edge set of $G(n,r)$ has law μ_r on X. By Theorem 3.1 and monotonicity, with probability at least $2/3$ it contains a path union, which is a copy of the edge set H; by the first remark of the Conventions, $G(n,r)$ then contains the original graph. As $2/3 \geq 1/2$, $p_c(n,H) \leq r$, which is (1.1). Finally $h \leq \binom{n}{2} \leq n^2$ and $\log_2 n \geq 1$ give $1 + \log_2 h \leq 1 + 2\log_2 n \leq 3\log_2 n$, and $3 \cdot 2048 = 6144$. □

Verification checklist

No step is marked unverified; the context paragraphs of Section 1 were not checked against the cited papers. A sceptical reader should check these first:

1. **Cancellation and independence in Lemma 2.1** (Steps 3, 4, 6): $\rho^{-|A_b|}\rho^{|A_b \setminus A_c|} = \rho^{-|A_c \cap A_b|}$, and the indicator of $A_b \setminus A_c \subseteq W$ is independent of any function of $W \setminus A_b$.
2. D_b **ignores** $A_v(b)$ (Lemma 3.2, Step 3). This is the only place where disjointness along paths enters the probability estimates; it licenses (2.7) with $f = D_b$ although one W serves all nodes.
3. **Exponent bookkeeping** (Lemma 3.2, Steps 4–5): $\mathrm{e}^m \cdot \mathrm{e}^{-9m}$ and $\mathrm{e}^m \cdot \mathrm{e}^{11m}$ in (3.5), then $12/15 - 8 \leq -7$ and $\sum_{m \geq 1} \mathrm{e}^{-7m} < 1/4$.
4. **Spread after conditioning** (Lemma 3.2, Step 7): it rests on $T \subseteq A_v(c)$ and on the source having law ν_v for every fixed W; the cost is only the factor $1/\gamma_v \leq 1/(1 - \mathrm{e}^{-m_j})$.
5. **Hypotheses persist under iteration** (Theorem 3.1, Steps 2–4): trimming keeps (3.7), the ratio 16 survives the floors, and the accumulated spread factor is below 4 because the capacities at one level are distinct integers.
6. **Hierarchy** (Lemma 4.1, Steps 1 and 5): $2 \cdot 17\mathrm{e}/128 < 1$ forces $|H_{i-1}| < |H_i|/17$; the spread bound uses that (4.2) depends only on isomorphism types, so maximality applies to $U \cong S_0 \cup J$.
7. **Constants**: $\rho = 4\mathrm{e}^{50}\sigma$, at most $4s$ sets, $\sigma = 128q$, giving $4 \cdot 4 \cdot 128 = 2048$; then $3 \cdot 2048 = 6144$.

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